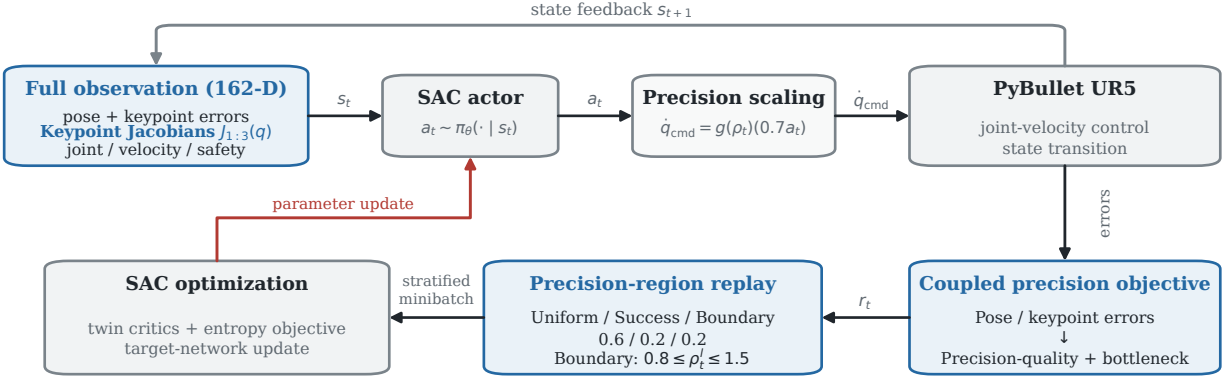
state feedback s_{t+1}
Full observation (162-D)
pose + keypoint errors
Keypoint Jacobians $J_{1:3}(q)$
joint / velocity / safety
s_t
SAC actor
$a_t \sim \pi_\theta(\cdot \mid s_t)$
a_t
Precision scaling
$\dot{q}_{\text{cmd}} = g(\rho_t)(0.7a_t)$
$\dot{q}_{\text{cmd}}$
PyBullet UR5
joint-velocity control
state transition
errors
parameter update
SAC optimization
twin critics + entropy objective
target-network update
stratified
minibatch
Precision-region replay
Uniform / Success / Boundary
0.6 / 0.2 / 0.2
Boundary: $0.8 \le \rho_t' \le 1.5$
r_t
Coupled precision objective
Pose / keypoint errors
↓
Precision-quality + bottleneck